

INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

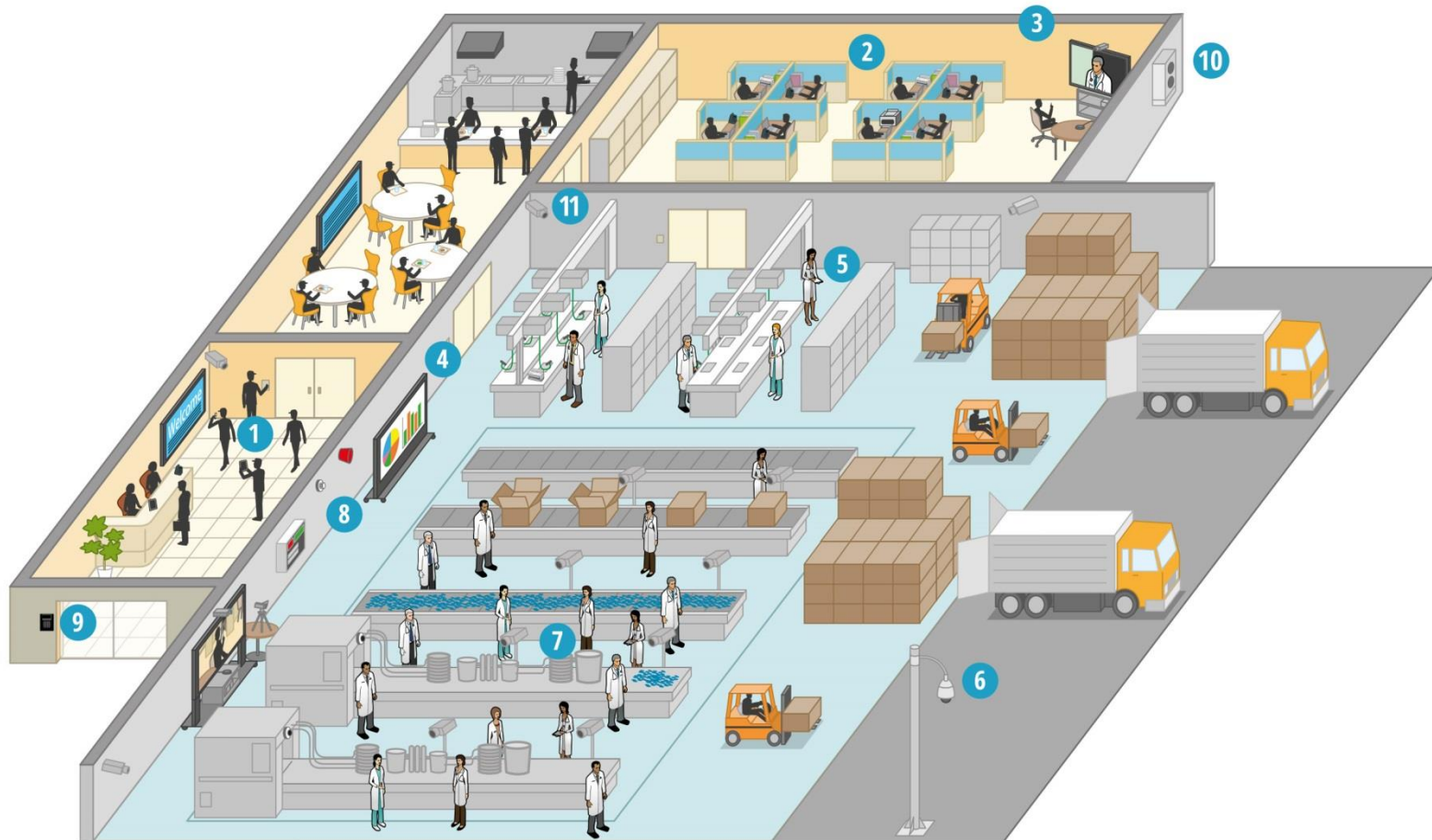
INDU 211

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Facility Layout

- Internal location problems
- Strategic, tactical or operational decisions
- Systematic layout solution
- Template
- Computer simulation
- Optimization

Facility layout design



The Need for Layout Decisions

Inefficient operations

High Cost
Bottlenecks

Changes in the design
of products or services

The introduction of new
products or services

Accidents
Safety hazards

The Need for Layout Decisions

Changes in
environmental
or other legal
requirements

Changes in volume of
output or mix of
products

Changes in methods
and equipment

Morale problems

Facility layout decision

- Location of areas & impacts on system flow
- System productivity and costs affected
- Cost of layout:
 - Construction
 - Installation
 - Material handling
 - Ease of future expansion
 - Production
 - Machine downtime
 - In-process storage
 - Safety
 - Ease of supervision
- Layout alternatives are limited by:
 - amount/type of space required/available

Facility layout design

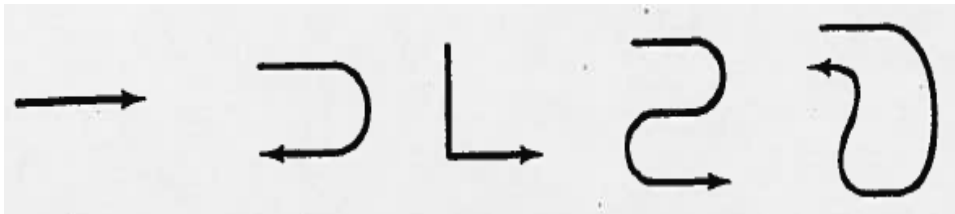
- Focus of IE:
 - Material handling cost (30 to 95% of production cost)
 - Variable cost
 - Quantifiable
- Simultaneous design of layout and material handling system
- Layout and resource management (energy, water, compressed air, etc.)
- Work environment conditions: temperature, humidity, air flow, corridor, safety signs, parking, washroom, cafeteria, etc.

Basic layout types

- Product Layout
- Process Layout
- Cellular Layout
- Fixed Position Layout
- Mixed Layout

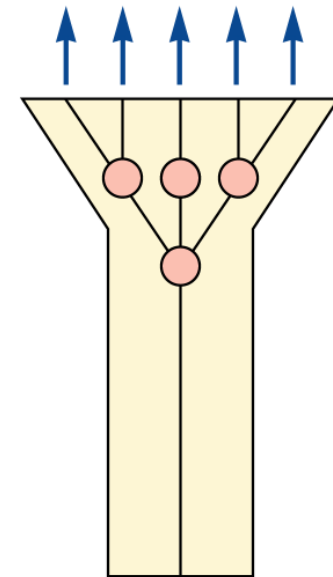
Product layout

- Mass production
 - Large volume of one or similar products
 - Continuous flow of material
 - Paper mills
 - Dairy products
 - Cement factories
 - Automotive assembly plants
 - Other names
 - Line flow production
 - Continuous production



Product-focused (continuous process)

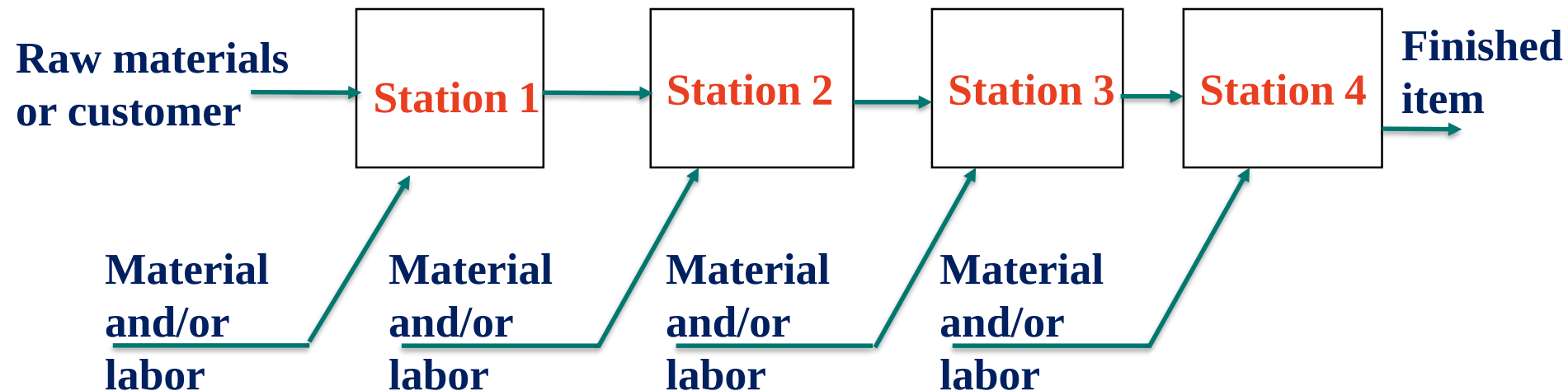
Output variations in size,
shape, and packaging



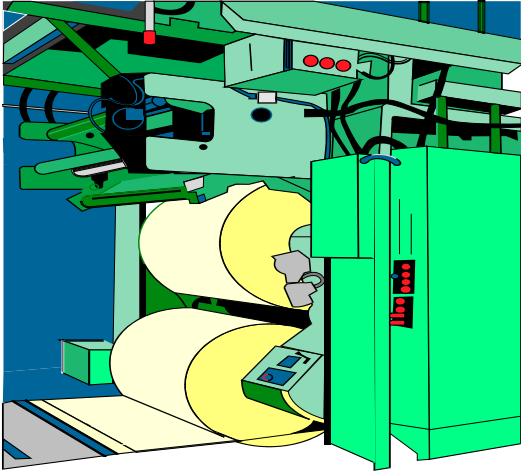
Few inputs

Product layout

- Operations arranged in required sequence
- Narrow product variety in high volumes
- Standardized operations for smooth and rapid flow



Product layout examples



© 1984-1994 T/Maker Co.

Paper
(Continuous)



Soft Drinks
**(Continuous,
then Discrete)**



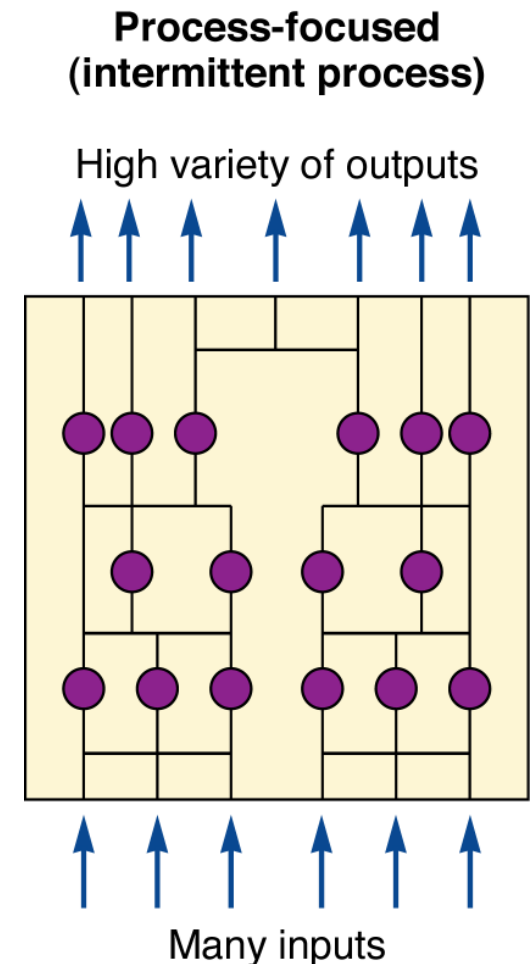
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Advantages and limitations of product layout

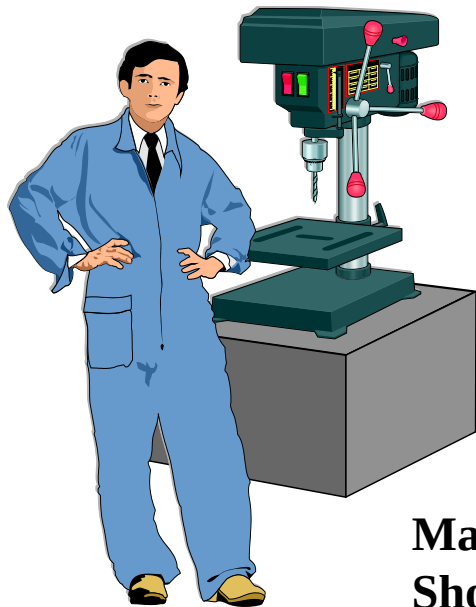
- Advantages:
 - Smooth flow lines
 - Small in-process inventory
 - Reduced production time and material handling (machines located in order)
 - Little operator skill
- Limitations:
 - Breakdown of one machine → complete stoppage of the line
 - Less flexible in terms of changing the product
 - Pace of production → slowest machine
 - High investment (identical machines, some of which not fully utilized)

Process layout

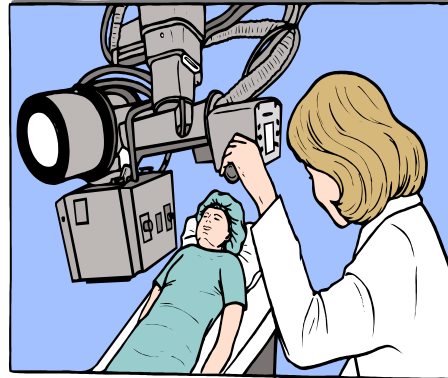
- Similar processes are together, equipment that perform similar processes are grouped together
- Job shop manufacturing (lathes in one area, drills in another area)
- Production flow varies according to the particular items being produced
- Small volume of many different products
- Other names
 - Intermittent process
 - Job shop



Process layout examples



**Machine
Shop**



Hospital

Bank



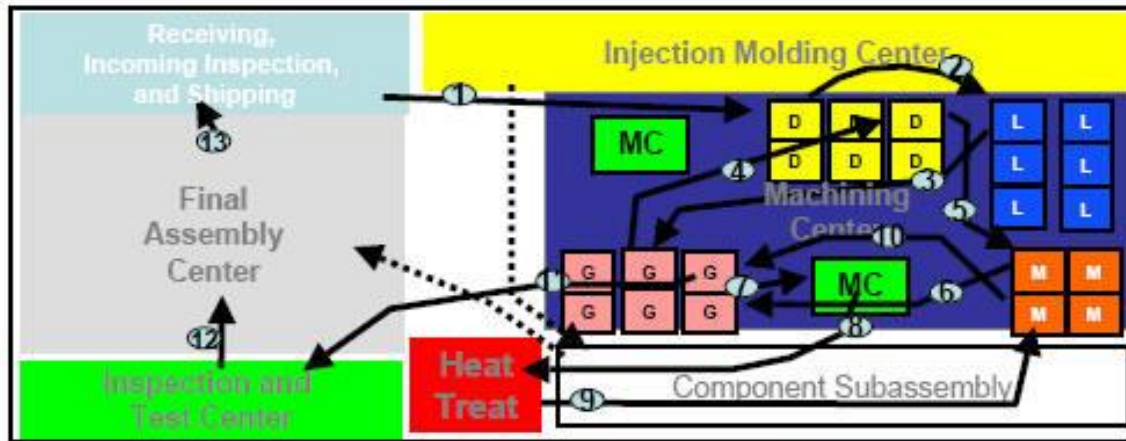
Advantages and limitations of process layout

- Advantages:
 - Better utilization of machines
 - High degree of flexibility
 - Lower investment in machines
 - Higher satisfaction of operators
- Limitations:
 - More expensive material handling
 - More difficult production planning and control
 - Large amounts of work-in-process inventory → more capital tied-up
 - Higher grades of skill

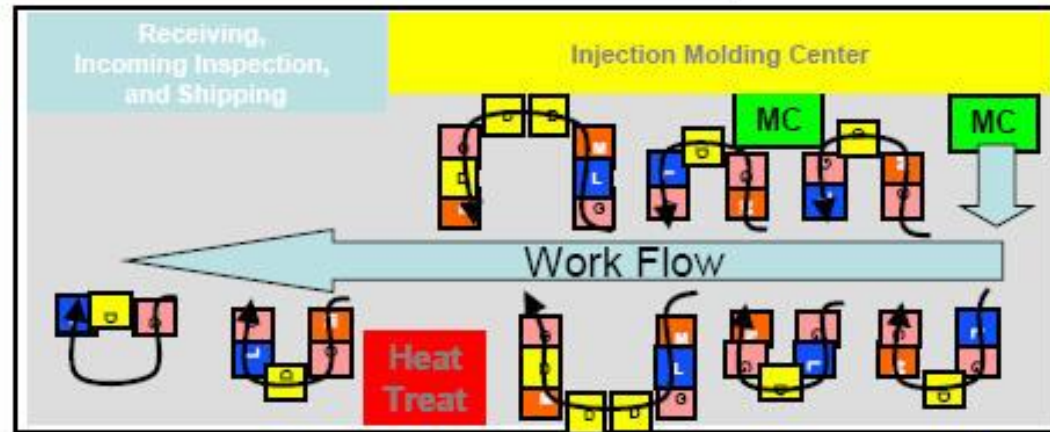
Cellular Layout

- “Group Technology”
- Machines are grouped in machine “cells”
- Suitable for mass customization (many product variants that share similar processing steps)
- Each cell is designed to process a group of similar parts
 - Dining chairs, office chairs, bar stools cells (furniture manufacturing)
 - Gear manufacturing and brake manufacturing cells (Automobile manufacturing)
- Smooth material flow within cells
- Minimizing inter-cell material flow
- More flexible (less efficient) than product layout
- More efficient (less flexible) than process layout

Process Layout vs Cellular Layout



Desired State



Fixed Position Layout

- Product stays in fixed position: personnel, material and equipment come to it
- Used when the product is very bulky, large, heavy or fragile



Mixed layout

- Automobile Manufacturing
 - Assembly line (product layout) for the main vehicle body
 - Process layout for support shops (welding, machining, painting, tool & die)
 - Cellular layout in subassembly areas (e.g., seat manufacturing cells)
- Electronics (e.g., smartphone manufacturing)
 - Product layout for PCB assembly lines
 - Process layout for specialized tasks (testing, repair, prototyping)
 - Cellular layout for final assembly & customization stations